

Supplementary Materials for C²GES: A Diagnostic Evaluation of Role-Conditioned Extractive Summarization for Long Power-System Technical Reports

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Contents and Data Files

This supplement separates scientific interpretation from implementation records. It contains thirteen tables and one figure referenced by the main manuscript. The main-text evidence-layer table (“Evidence layers and claim bounds”) states the four evaluation layers and what each may and may not support; the tables below are implementation and exploratory-component records for those layers, not a confirmatory external test. Paths below are relative to the C2GES release root. Machine-readable files are provided under `03_Reproducibility/Data/`: the non-verbatim sampling frames, aggregate metrics, paired ROUGE-L differences, output-length records, exact sign-flip results, page locators, component-factorial diagnostics, and development-only calibration report. Source PDFs and verbatim extracted passages are excluded because third-party redistribution permission has not been established.

Table S1. Rights-Safe Sampling Frame

The complete 40-row sampling frame is supplied as `03_Reproducibility/Data/rights_safe_metadata/rights_safe_report_metadata.csv`. Its 23 fields include the report identifier, source locator, split, series identifier, inclusion decision, source-file hash, and redistribution status; it contains no report passages.

Quantity	Value
Inspected complete PDFs	40
Declared pages	3200
Retained / excluded reports	27 / 13
Development / test reports	12 / 15
Non-summary candidates	12,924

Table S2. Output-Length Records

The complete 14 condition-budget summary is supplied as `03_Reproducibility/Data/postrun_diagnostics/output_length_summary.csv`; per-report values are in `03_Reproducibility/Data/postrun_diagnostics/output_length_per_report.csv`. “Unrenorm. removal” denotes the unrenormalized coefficient-removal comparison.

<i>K</i>	Condition	Mean words	Mean chars.	> 100	Table	Max. words
5	Full C ² GES	287.7	1843.6	12	14	140
5	Unrenorm. removal	329.0	2099.9	15	16	270
5	Semantic-MMR	184.7	1242.1	5	9	138
5	TextRank	177.0	1125.1	1	7	124
10	Full C ² GES	568.9	3619.5	25	26	270
10	Unrenorm. removal	596.9	3822.5	28	26	270
10	Semantic-MMR	354.5	2361.2	7	16	138
10	TextRank	369.0	2329.7	4	18	130

Table S3. Corpus and Implementation Checks

Checked property	Recorded result
Reference/body page-interval overlap	0
Normalized exact candidate/reference matches	0
Normalized common substrings of at least 50 characters	0
Report-condition-budget records / expected records	210 / 210
Selected source references resolved to a valid page	1575 / 1575
Maximum path expansions / configured limit	19,638 / 2,000,000
Maximum qualifying paths / configured limit	16,182 / 250,000
Full/unrenormalized-removal score comparisons changed	9774 / 19,008
Report-budget selection sequences changed	28 / 30
Aggregate and paired-contrast recomputation	Exact agreement

Aggregate values are available at `03_Reproducibility/Data/audits/aggregate_metrics.json`; non-verbatim paired differences and page locators are available at `03_Reproducibility/Data/figure_inputs/paired_rougel_differences_nonverbatim.csv` and `03_Reproducibility/Data/postrun_diagnostics/selected_page_locator.csv`. Detailed checks are retained with the corresponding code and archived run records. Passing these checks establishes consistency of the reported computation; it does not determine semantic summary quality.

Table S4. Exact Paired Sign-Flip Sensitivity

For each contrast, all $2^{15} = 32,768$ report-level sign assignments were enumerated. The procedure assumes report-level independence and sign exchangeability. Holm adjustment treats the six values as one family.

K	Baseline	Signs + / - / 0	Exact p	Holm-adjusted p
5	Unrenorm. removal	7/7/1	0.436768	0.436768
5	Semantic-MMR	14/1/0	0.000305	0.001526
5	TextRank	14/1/0	0.000122	0.000732
10	Unrenorm. removal	6/8/1	0.200684	0.401367
10	Semantic-MMR	11/4/0	0.006592	0.026367
10	TextRank	13/2/0	0.009644	0.028931

These post-result values are sensitivity analyses. The main manuscript emphasizes effect sizes, paired distributions, and report-composition intervals.

The complete machine-readable result is `03_Reproducibility/Data/postrun_sensitivity/exact_signflip_results.csv`; the JSON file in the same directory retains the report-level non-verbatim deltas used in the calculation.

Table S5. Post-Access External-Series Inventory

The fixed exploratory inventory contained eight public power-system reports from four organizations. Seven PDFs were acquired and analyzed; the NESO item returned HTTP 403 and was retained as a documented acquisition failure rather than replaced. Metadata or search snippets had been seen before the protocol was frozen, so all rows are classified as exploratory and none permits a confirmatory claim.

Organization	Fixed	Analyzed	Evidence status
ENTSO-E	3	3	Post-access exploratory
NERC	3	3	Post-access exploratory
NESO	1	0	Acquisition failed (HTTP 403)
Transpower	1	1	Post-access exploratory
Total	8	7	Confirmatory claims disallowed

The non-verbatim record, source pages, direct-download locators, access date, and redistribution status are supplied in `03_Reproducibility/Data/exploratory_external_v0/EXPLORATORY_EXTERNAL_INVENTORY.csv`.

Table S6. Controlled Comparator and Budget Conditions

Control	Frozen implementation	Diagnostic purpose
Candidate pool	One layout-filtered pool per report for every method	Removes candidate-access asymmetry
Output budget	Complete-ranking greedy selection at 110 and 260 words; no sentence truncation	Matches emitted information quantity
Development opportunity	Nine configurations for each tunable system in the exploratory registry	Balances search opportunity
Series unit	Seven reports in seven distinct series	Supports equal-series contrasts and LOSO checks
Component scale	Positive weights renormalized to sum to one	Separates component removal from score-scale change
Failure policy	Every fixed item retained; execution status saved per cell	Prevents silent replacement or deletion

The v3 run completed all 182 condition–report–budget cells. These controls were fixed after source access and therefore improve internal comparability without converting the pilot into a prospective confirmatory experiment.

Table S7. Principal Component Contrasts

Family	Contrast and budget	Mean Δ R-L	95% cluster interval	Exact p	Holm p
Incremental	AB-5 – AB-6, 110	-0.00799	[-0.01949, 0.00149]	0.34375	1.00000
Incremental	AB-5 – AB-6, 260	-0.00718	[-0.01855, 0.00151]	0.28125	1.00000
Graph type	G-T – G-U, 110	0.00139	[-0.01481, 0.01959]	0.90625	0.90625
Graph type	G-T – G-U, 260	-0.00544	[-0.01640, 0.00594]	0.35938	0.71875
Factorial	Reservation main, 110	-0.00971	[-0.01751, -0.00244]	0.04688	0.23438
Factorial	Path main, 110	-0.00247	[-0.01787, 0.01333]	0.79688	1.00000
Factorial	Interaction, 110	-0.01104	[-0.03173, 0.00264]	0.34375	1.00000
Factorial	Reservation main, 260	0.00047	[-0.00387, 0.00530]	0.87500	1.00000
Factorial	Path main, 260	-0.01034	[-0.01871, -0.00371]	0.03125	0.18750
Factorial	Interaction, 260	0.00632	[-0.00222, 0.01602]	0.25000	1.00000

No component contrast survived its separately defined Holm family. Intervals use series-cluster resampling, and exact values enumerate the 2^7 series sign assignments. Full series effects, standardized paired effects, and leave-one-series-out estimates are provided in `factorial_series_effects.csv` and `factorial_loso.csv` under the v3 factorial directory.

Table S8. Complete AB-0–AB-6 Component Chain

Budget	Condition	R-L	Redundancy	Role cov.	Edge cov.	Words/util.	Time (s)/MB
110	AB-0	0.1564	0.2308	0.286	0.662	108.4/0.986	0.269/1.04
110	AB-1	0.1582	0.1527	0.524	0.869	108.4/0.986	0.268/1.04
110	AB-2	0.1696	0.0783	1.000	0.890	108.1/0.983	0.253/1.03
110	AB-3	0.1707	0.0800	1.000	0.971	108.1/0.983	0.248/1.03
110	AB-4	0.1577	0.0564	1.000	0.964	109.0/0.991	0.249/1.03
110	AB-5	0.1588	0.0508	1.000	0.964	108.6/0.987	0.252/1.06
110	AB-6	0.1668	0.0595	1.000	0.948	108.4/0.986	0.252/1.08
260	AB-0	0.1687	0.1799	0.429	0.626	258.0/0.992	0.350/1.12
260	AB-1	0.1873	0.1185	0.714	0.885	259.3/0.997	0.347/1.12
260	AB-2	0.1843	0.1128	1.000	0.906	258.6/0.995	0.305/1.12
260	AB-3	0.1848	0.1038	1.000	0.971	257.7/0.991	0.314/1.12
260	AB-4	0.1805	0.0915	1.000	1.000	257.9/0.992	0.313/1.12
260	AB-5	0.1772	0.0509	1.000	1.000	258.1/0.993	0.344/1.45
260	AB-6	0.1844	0.0747	1.000	0.984	257.4/0.990	0.342/1.46

All failure rates were zero. Runtime is the recorded per-report estimate and memory is Python peak allocation; the separately loaded embedding model is excluded. Role and edge coverage measure heuristic formal coverage, not human-validated semantic correctness.

Table S9. Adjacent-Corpus Construct Audit of Role and Edge Cues

Two public corpora carrying human annotation were used to bound what the lexical role and edge channels can carry. The deterministic role rule and its transition rule were applied unchanged; neither corpus is redistributed, and both must be obtained from their original distributors. This is an adjacent-corpus construct audit, not the domain expert adjudication reserved in the main text.

Corpus	Endpoint	Units	Human positive rate	Precision	Recall	Note
GUM eRST	Dominant role assigned	7678 EDUs	—	—	—	349 units (4.55%)
GUM eRST	Admissible typed edge	7285 pairs	0.179	not estimable	not estimable	45 pairs (0.62%) had role evidence on both units
EBM-NLP	interventions → mitigation	2139 sentences	0.524	0.610	0.064	382 sentences (17.9%) received a dominant role
EBM-NLP	outcomes → impact	2139 sentences	0.519	0.807	0.060	same sentence set as above

The GUM rows use the human enhanced-RST relation labels of the Georgetown University Multilayer corpus; the EBM-NLP rows use test spans annotated by medical professionals, with two roles mapped onto C2GES role names (intervention to mitigation, outcome to impact) and the participant label reported as unmapped. The cues are precise when they fire and sparse and genre-bound in general text; neither row closes any precondition of the annotated-study gate in the main text, and no row supports a role, edge, path, faithfulness, or omission claim.

Table S10. External Prospective Evaluation on a New Report Family

The inventory, reference rule, budgets, comparators, endpoints and statistics were frozen before any score was computed (protocol and amendments in 03_Reproducibility/Data/external_prospective_v1/). The corpus was acquired on 22–23 September 2026 from the public document stores of ENTSO-E and from additional NERC reports; it is a frozen external corpus, not an author-attested unseen set, and it is never pooled with the historical 15-report test or the seven-series pilot.

Selector	n	Mean ROUGE-L @110	Mean ROUGE-L @260	Word budget
no-path C ² GES	19	0.125803	0.141786	110 / 260
Full C ² GES (path 0.10)	19	0.123381	0.138880	110 / 260
TextRank	19	0.149076	0.164726	110 / 260

Contrast	Budget	Mean Δ R-L	Bootstrap 95%	Exact p	Holm p	Negative / positive documents
Full – no-path	110	-0.002422	[-0.005913, 0.000310]	0.164062	0.317284	5 / 3
Full – no-path	260	-0.002906	[-0.005605, -0.000636]	0.015625	0.046875	7 / 0
TextRank – no-path	110	0.023273	[-0.005924, 0.055280]	0.158642	0.317284	6 / 13
TextRank – no-path	260	0.022940	[0.008540, 0.038543]	0.008095	0.032380	5 / 14

Composition: six ENTSO-E grid incidents (including the 2024 South-East Europe and 2025 Iberia reports), seven ENTSO-E market-coupling incidents, and six NERC reports; 21–2,000 candidate units and 102–10,935 reference words per document. References are the official summary sections detected after converting each PDF to structured Markdown; the detector accepts Executive Summary, Management Summary and Summary headings, which is a documented generalisation of the earlier Executive-Summary rule made before scoring. Holm adjustment covers the four contrasts shown. The two corrected contrasts favour the untyped ranking over the typed path channel; they do not establish system superiority, expert construct validity, or operational benefit, and they are not author-attested unseen evidence.

Table S11. Error Profile of Path-Driven Selection Changes

Descriptive decomposition of the units whose selection changed between Full (AB-5) and no-path (AB-6) in the seven-series matched-budget pilot, grouped by the sign of the report-level ROUGE-L difference. Changed units are the symmetric difference of the two selection sets in a report–budget cell; cells are counted when at least one unit changed. “Blocked pairs” count ordered role-compatible pairs (a permitted stage-monotone transition) whose positions differ by more than the 12-position window and therefore could not form an edge. Chronology-violating incident edges are typed edges whose source unit occurs later in the document than its target; direction follows the role stage, not document order, so a high share is a structural property of the design rather than an error count. Machine-readable per-unit records and inputs are in `03_Reproducibility/Data/descriptive_addenda_v1/`. This profile describes where the path channel acts; it does not apportion causality for the endpoint difference.

Budget	Δ R-L sign	Cells	Changed units	Edge-incident	Chronology-violating incident	Mean max incident distance	Mean blocked pairs
110	negative	4	21	0.952	0.800	9.9	83.3
110	zero	1	5	1.000	1.000	12.0	417.8
110	positive	2	11	0.818	1.000	9.9	33.4
110	all	7	37	0.919	0.882	10.2	113.7
260	negative	4	27	0.963	0.885	10.5	50.0
260	positive	2	22	1.000	0.909	11.5	362.8
260	all	6	49	0.980	0.896	11.0	190.4

At both budgets the units the path channel moves are almost always edge-incident (92–98%), and most of those touch at least one chronology-violating edge (80–100%). The endpoint-negative cells do not show an exceptional profile: at 110 words they exhibit fewer blocked beyond-window pairs per changed unit than the zero/positive cells, and at 260 words fewer than the positive cells. The 12-position window leaves a large stock of role-compatible pairs unreachable (tens to hundreds per changed unit), so long-range dependencies remain untested by design rather than adjudicated by evidence.

Table S12. Per-Role Token Alignment Between Extracts and References

Descriptive cue-versus-cue alignment for the role layer in the same pilot. Both sides are labeled by the pipeline’s deterministic lexical cues (unit-scaled within each text set, abstaining on ties): a selected unit “has” a role when its dominant role equals that label, and reference sentences are labeled the same way after a deterministic sentence split. Token precision is the share of stopword-filtered content tokens of the role- r selected units that also occur in role- r reference sentences; token coverage is the share of role- r reference-sentence tokens recovered anywhere in the extract. Sentence-level ROUGE-L matching was evaluated first and essentially never fires, because executive-summary sentences are longer and differently worded than source units; token alignment is the informative granularity. Reference-side roles are sparse (root cause never fires on any of the seven references), so empty cells are structural, not missing data.

These are alignment measures under the cue definition, not expert-validated correctness, and they close no row of the evidence gates.

Role	Ref. reports	110-word budget				260-word budget			
		Precision		Coverage		Precision		Coverage	
		Full	no-path	Full	no-path	Full	no-path	Full	no-path
Root cause	0/7	0	0	—	—	0	0	—	—
Trigger event	1/7	0	0	0.053	0.053	0.008	0	0.105	0.158
Propagation or response	4/7	0.251	0.106	0.217	0.243	0.213	0.084	0.397	0.369
Impact	4/7	0.093	0.075	0.144	0.072	0.050	0.040	0.252	0.249
Mitigation	2/7	0.024	0.061	0.125	0.188	0.061	0.061	0.225	0.275

Macro means over the seven reports; empty cells mark roles that never fire on that reference. Precision reads low in absolute terms because the same content is expressed with different words on the two sides; the relative pattern is the informative part. The role layer’s effect is concentrated where the cues and the references coincide at all (propagation/response, impact, mitigation), and the path channel does not systematically raise alignment: Full exceeds no-path in propagation precision at both budgets but trails it in mitigation precision at 110 words and in mitigation coverage at both budgets.

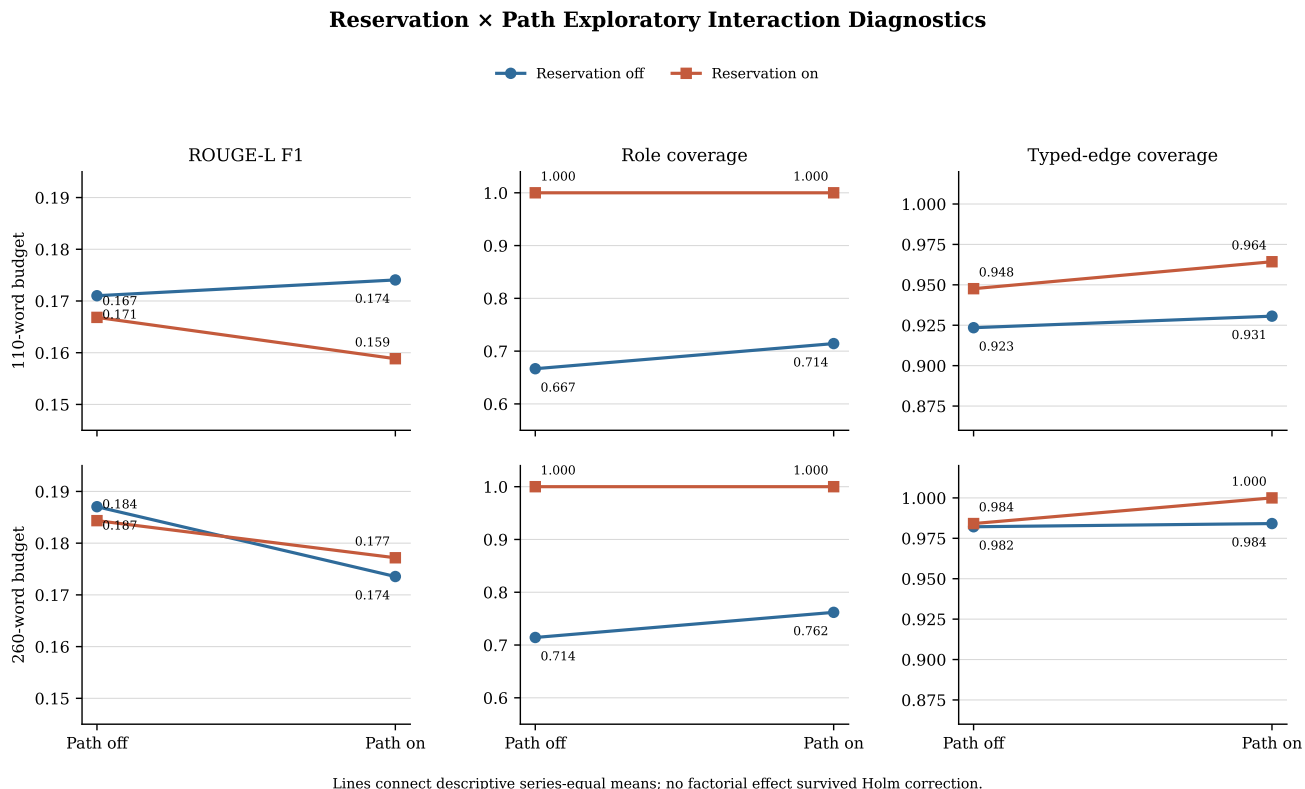
Table S13. Long-Block Audit Detail

Per-condition counts of long extraction units in the legacy candidate stream (unit budgets $K=5$ and $K=10$, 15 historical test reports; columns sum selection instances, so a unit selected under several conditions is counted in each) and the block-preserving re-segmentation audit over all 27 historical reports. The audit preserved page/block boundaries, excluded recurrent margin material, and dropped blocks overlapping detected tables; it was not used to rerun ranking, so whether the relative ordering at matched budgets would change remains unanswered by design.

Condition	$K=5$			$K=10$		
	Units >100 words	Table-marker units	Max unit words	Units >100 words	Table-marker units	Max unit words
Lead	5	2	119	6	7	119
Centroid	11	15	221	13	29	221
TextRank	1	7	124	4	18	130
Semantic-MMR	5	9	138	7	16	138
Role	3	6	270	6	12	270
Graph, no path (strict)	15	16	270	28	26	270
Full C ² GES	12	14	140	25	26	270

Under the block-preserving rule the same reports yield 14,290 layout units of which 39 exceed 100 words (spread over 16 of 27 reports); the per-report maximum unit length has median 103 and worst case 522 words. Non-verbatim examples of long legacy selections are recorded in `q10_long_unit_examples_nonverbatim.csv` within the same data directory: the longest Full selection is a 270-word unit sharing 21 content tokens with its reference summary, and a 162-word Full selection shares 40. Long units can therefore carry substantial reference-token mass into an extract, which is the mechanism behind the output-length diagnostic in the main text; the audit quantifies the exposure but does not license re-ranking claims.

Figure S4. Reservation–Path Factorial Interaction



Lines compare reservation off/on while the horizontal axis switches the normalized path term. The rows show 110- and 260-word budgets; columns show ROUGE-L, formal role coverage, and formal typed-edge coverage. The budget-dependent ROUGE-L patterns do not establish a stable interaction after family-wise correction.

Exploratory Development-Only Calibration

The complete chronology, configuration space, leave-one-report-out results, and machine-readable records are described in `03_Reproducibility/Data/POST_UNBLINDING_DEV_CALIBRATION_SUPPLEMENT.md`. A zero path-deletion weight was selected in all 12 development folds; the analysis was not applied to the retained 15-report test set.

A separate equal-budget exercise is documented in `03_Reproducibility/Data/dev_balanced_tuning/`. Semantic-MMR, TextRank, and normalized-path C²GES each received nine configurations and the same ordered development objective over both extraction budgets. The selected values were 0.9, 0.65, and 0.0, respectively. The decision record states `test_input_accessed=false`; these configurations are authorized only for a future preregistered external-series comparison and do not alter the retained-test results.